

Thm. Let $\{a_n\}$ be a sequence of real numbers, and let $S_k \stackrel{\text{def}}{=} a_1 + \dots + a_k$ ($1 \leq k \in \mathbb{N}$).
 If $\lim_{n \rightarrow \infty} a_n = A \in \mathbb{R}$, then $\lim_{n \rightarrow \infty} \frac{S_n}{n} = A$.

Proof. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow |a_n - A| < \varepsilon$.
 Meanwhile, observe that

$$\left| \frac{S_n}{n} - A \right| = \frac{1}{n} \cdot |a_1 + \dots + a_n - nA| = \frac{1}{n} \cdot |a_1 - A + a_2 - A + \dots + a_n - A|$$

Now, for all $n \geq N$,

$$\left| \frac{S_n}{n} - A \right| = \frac{1}{n} \cdot |a_1 - A + \dots + a_N - A + a_{N+1} - A + \dots + a_n - A|$$

$$\leq \frac{1}{n} \cdot (|a_1 - A| + |a_2 - A| + \dots + |a_n - A|)$$

$$= \frac{1}{n} \cdot \left(\sum_{i=1}^N |a_i - A| \right) + \frac{1}{n} \cdot \left(\sum_{i=N+1}^n |a_i - A| \right)$$

$$\leq \frac{1}{n} \cdot \left(\sum_{i=1}^N |a_i - A| \right) + \frac{1}{n} \cdot (n - N) \cdot \varepsilon$$

$$\leq \frac{1}{n} \cdot \left(\sum_{i=1}^N |a_i - A| \right) + \varepsilon$$

Since N was fixed, $\limsup_{n \rightarrow \infty} \left| \frac{S_n}{n} - A \right| \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \cdot \left(\sum_{i=1}^N |a_i - A| \right) + \varepsilon = \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\limsup_{n \rightarrow \infty} \left| \frac{S_n}{n} - A \right| = 0$, so $\lim_{n \rightarrow \infty} \frac{S_n}{n} = A$. $\square$

This theorem is called the Cesaro mean,
 and in the next page, we deal with the generalization which is called the
 Stolz-Cesaro theorem.

Stolz-Cesaro Theorem.

Let $\{a_n\}$ be a sequence of real numbers,
and $\{b_n\}$ be a strictly increasing which diverges to $+\infty$.

$$\text{Then, } \liminf_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} \stackrel{①}{\leq} \liminf_{n \rightarrow \infty} \frac{a_n}{b_n} \stackrel{②}{\leq} \limsup_{n \rightarrow \infty} \frac{a_n}{b_n} \stackrel{③}{\leq} \limsup_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n}.$$

Moreover, if the limit $\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n}$ exists,

$$\text{then } \lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

Proof. To justify the inequalities, enough to show that the ①, because replacing $\{a_n\}$ to $\{-a_n\}$ in ①, we obtain ③ directly, and ② is given by definition.

If the first term in the inequality is $-\infty$, then there is nothing to prove.

Assume that the term has finite value.

$$\text{Suppose that } \alpha \in \mathbb{R} \text{ satisfies } \alpha < \liminf_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n}.$$

Then, there exists $N \in \mathbb{N}$ such that

$$k \geq N \Rightarrow \alpha < \frac{a_{k+1} - a_k}{b_{k+1} - b_k} \Rightarrow \alpha(b_{k+1} - b_k) < a_{k+1} - a_k.$$

For $m \geq N$, by the telescoping,

$$\alpha \cdot [b_m - b_N] = \alpha \cdot \sum_{i=N+1}^m (b_i - b_{i-1}) < \sum_{i=N+1}^m (a_i - a_{i-1}) = a_m - a_N$$

dividing $b_m - b_N > 0$ to the both side, then

$$\alpha < \frac{a_m - a_N}{b_m - b_N} = \frac{a_m}{b_m - b_N} - \frac{a_N}{b_m - b_N} = \frac{a_m}{b_m} \cdot \frac{1}{1 - b_N/b_m} - \frac{a_N}{b_m - b_N}$$

Now, since b_m diverges to $+\infty$, taking $\liminf_{m \rightarrow \infty}$ to the both side, then

$$\alpha \leq \liminf_{m \rightarrow \infty} \frac{a_m}{b_m}.$$

Consequently,

$$\alpha < \liminf_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} \Rightarrow \alpha \leq \liminf_{n \rightarrow \infty} \frac{a_n}{b_n} \text{ implies } \liminf_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} \leq \liminf_{n \rightarrow \infty} \frac{a_n}{b_n}.$$

(To clarify, consider the negative). Further, the base assertion is obtained by the squeeze theorem. $\square$

Corollary,

Let $\{x_n\}$ be a sequence of real numbers.

If $\lim_{n \rightarrow \infty} x_n = L$, then $\lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \dots + x_n}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \left[\sum_{i=1}^n x_i \right] = L$.

Proof. Let $a_n = \sum_{k=1}^n x_k$ and $b_n = n$, for each $n \in \mathbb{N}$.

Then clearly $b_n \rightarrow \infty$ and strictly increase, so by the Stolz-Cesaro Theorem,

$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \lim_{n \rightarrow \infty} x_{n+1} = L \text{ implies that } \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{x_1 + \dots + x_n}{n} = L. \quad \square$$

Corollary,

Let $\{x_n\}$ be a sequence of positive real numbers.

If $\lim_{n \rightarrow \infty} x_n = L > 0$, then $\lim_{n \rightarrow \infty} \sqrt[n]{x_1 x_2 \dots x_n} = \lim_{n \rightarrow \infty} \left[\prod_{i=1}^n x_i \right]^{\frac{1}{n}} = L$.

Proof. Let $y_n = \ln(x_n)$ for each $n \in \mathbb{N}$, then by the above corollary,

$$\lim_{n \rightarrow \infty} \frac{y_1 + \dots + y_n}{n} = \lim_{n \rightarrow \infty} \ln \left(\sqrt[n]{x_1 \dots x_n} \right) = \ln L,$$

(Since the $\ln$ function is continuous, it preserves the convergence) so proved. $\square$

Corollary,

Let $\{x_n\}$ be a sequence of real numbers, and $\{y_n\}$ be a positive sequence satisfying

$$\sum_{i=1}^{\infty} y_i = \infty.$$

If $\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = L$, then $\lim_{n \rightarrow \infty} \frac{x_1 + \dots + x_n}{y_1 + \dots + y_n} = L$.

Proof.

Let $a_n = \sum_{k=1}^n x_k$ and $b_n = \sum_{k=1}^n y_k$ for each $n \in \mathbb{N}$. Since $\{y_k\}$ is positive and the sum diverges to $+\infty$, therefore it satisfies the condition of Cesaro theorem, with $\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \lim_{n \rightarrow \infty} \frac{x_{n+1}}{y_{n+1}} = L$, so

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{x_1 + \dots + x_n}{y_1 + \dots + y_n} = L. \quad \square$$